

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.1 Worksheet

1. Consider the points $A(-4, 0, -1)$, $B(3, 1, -5)$, & $C(2, 4, 6)$.

a) Determine the distance between points A and B.

b) Which of these points is closest to the yz -plane?

c) Which point lies in the xz -plane?

d) Which point is closest to the origin?

2. Find the distance from $(4, -2, 6)$ to each of the following.

a) The xy -plane

b) The yz -plane

c) The xz -plane

d) The x -axis

e) The y -axis

f) The z -axis

3. Find an equation of a sphere with center $(-3, 2, 5)$ and radius 4. What is the intersection of this sphere with the yz -plane?
4. Find an equation of the sphere that passes through the point $(4, 3, -1)$ and has center $(3, 8, 1)$
5. Find an equation of the sphere that passes through the origin and whose center is $(1, 2, 3)$
6. Show that the equation represents a sphere, and find its center and radius.
- a) $x^2 + y^2 + z^2 - 2x - 4y + 8z = 15$

b) $2x^2 + 2y^2 + 2z^2 = 8x - 24z + 1$

7. Prove that midpoint of the line segment from $P_1(x_1, y_1, z_1)$ to $P_2(x_2, y_2, z_2)$ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$

8. Find an equation of a sphere if one of its diameters has endpoints $(2, 1, 4)$ and $(4, 3, 10)$.

9. Find equations of spheres with center $(2, -3, 6)$ that touch (but do not intersect)

a) the xy -plane

b) the yz -plane

c) the xz -plane

10. What does the equation $x = 4$ represent in $\mathbb{R}^2$? What does it represent in $\mathbb{R}^3$? Provide sketches of each of these.

11. Describe in words the region of $\mathbb{R}^3$ represented by the following equations or inequalities.

a) $y < 8$

b) $0 \leq z \leq 6$

c) $x + y = 2$

d) All points where $x^2 + y^2 = 4$ & $z = -1$

e) $x^2 + y^2 + z^2 \leq 3$

f) $x^2 + z^2 \leq 9$

12. Create a sketch of the following cylindrical surfaces in $\mathbb{R}^3$.

a) $y = e^x$

b) $z = e^x$

c) $z = e^y$

d) $z = 1 - y^2$

e) $xy=1$

f) $z=\cos(y)$

g) $z=x+1$